Dashboard Learn Practice Compete

Pre Security > Computer Fundamentals > Inside a Computer System

Inside a Computer System

This room covers the basic components of a computer system.

45 min 248,509

Save Room | 1339 Recommend | Options

Room progress: 25%

Task 1 Introduction

Welcome to the first room of the Computer Fundamentals module! Before we can talk about security, we need to first understand what we are securing. Consider the following analogy: *Before protecting a castle, we need to know the layout of the castle: where the treasure room, the food storage, and the commander's quarters are. We need to know who enters and how they enter the castle. Also important is who can enter these rooms and take or put treasure or food, for example.*

The bottom line is: Trying to defend what you don't understand is like defending a castle you have never seen.
In this room, we will explore our "castle". We will cover what a computer is, its building blocks, and how they are connected. After completing this room, you will have a general idea of how the components of a computer system interact with each other to provide services to its users. Don't worry about too much technical jargon or depth; we'll take this nice and easy and focus on the fundamentals.

Learning Objectives

- After completing this room, you will be able to recognize and understand the functions of various computing components.

Prerequisites

Motivated to step into the digital world of computing

Answer the questions below

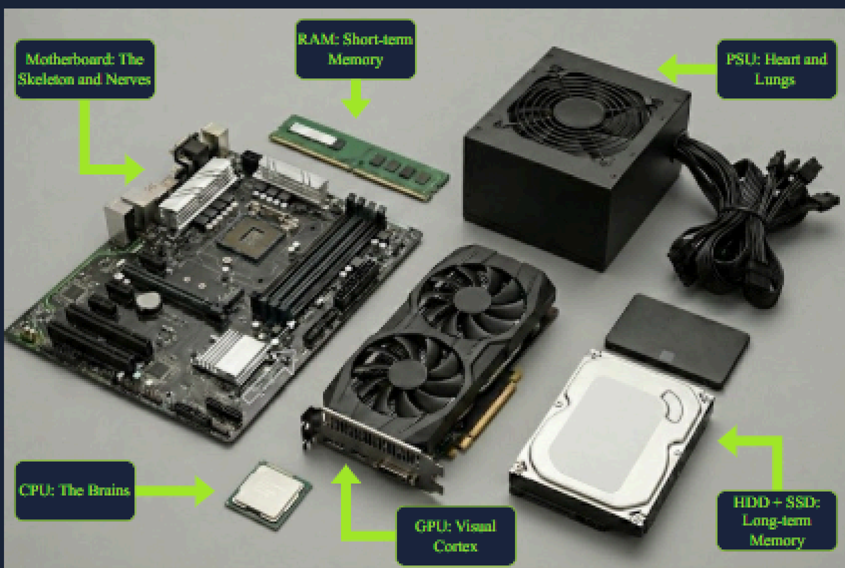
Let's get started!

No answer needed

✓ Correct Answer

Task 2 Inside a Computer System

Nearly every computer system that you can think of includes, in one way or another, the same building blocks. Each part has its own job, and together they make the computer work. Let's have a look at each of these building blocks.



Throughout this task, we will use a very relatable analogy: the human body. The image above shows the different PC components and its human counterpart. To learn more about each component, open the "Static Site" by clicking on the green "View Site" button below. The static site should open in split-screen view.

View Site

Answer the questions below

Give in the flag you received after completing the exercise on the static site.

THM{4llpccomp0n3nts1d3nt1f13d}

✓ Correct Answer

Computer Components

100%

1 Components ✓ 2 Inspection ✓ 3 Exercise ✓

Component Placement Exercise

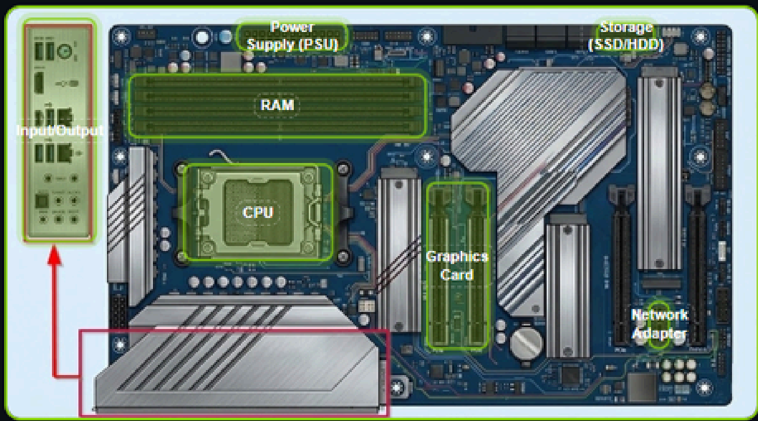
Drag each component to its correct connector on the motherboard. On mobile, tap a component to select it, then tap a slot to place it.

Back to Inspection

Verify

Components

Placed CPU	Placed RAM
Placed Graphics Card	Placed Network Adapter
Placed Storage (SSD/HDD)	Placed Power Supply (PSU)
Placed Input/Output	



Tip

Drag each component to the connector it belongs to — or tap a component to select it, then tap a slot to place it. You can move a component to a different slot at any time.

Boot Sequence

100%

1 Boot Sequence ✓ 2 Summary

Boot Sequence

Learn how a computer starts up, step by step. Complete the animation phase, then prove your knowledge with a drag-and-drop challenge.

Progress 6 / 6

1 Animation ✓ 2 Drag & Drop Challenge ✓

Challenge Completed!

You've mastered the boot sequence order.

Task 3 What Happens When You Press the Start Button?

Now that the core components are installed in the computer system, it is time to boot up the system. We can compare this to how we wake up in the morning and do a quick check to see if everything is working. Only when everything is OK, do we get up and start our day. The image below shows the steps a computer system goes through before it shows you a working interface (in the form of an Operating System).

Step 1: Press the Power Button
When we press the power button on our computer system, a signal is sent to the PSU to allow power to flow. Imagine our body being powered off when we sleep. Once we wake up and receive oxygen, our body starts pumping blood and boots up.

Step 2: Firmware starts
Continuing our analogy from step 1, once the body has started up, our core components are up and running, but our brain is not yet conscious. Like our bodies, a computer system contains firmware that allows all its components to start up. The central system that manages this is called the Unified Extensible Firmware Interface (UEFI). **Note: We will often see the term BIOS mentioned instead of UEFI. BIOS does the same as UEFI, but has mainly been replaced by UEFI**

Step 3: Power-On Self Test
Now that our body is up and running, it is time to test if everything is functioning as it should. If something isn't, there will be some alarm signals. One of the routines that the UEFI loads is the Power-On Self Test, which tests if every required component is present, configured correctly, and functioning.

Step 4: Select Boot Device
Once our body is up and running, configured correctly and fully functional, our system searches for the location of our bootup routine to start our consciousness. In our computer system the UEFI holds an ordered list which prioritizes on which device to look first for the boot up routine for the Operating System.

Step 5: Initiate Bootloader
Now that our system knows the part of our brain where our consciousness is located, it initiates the "load routine" to start it. Our computer systems follow a similar process: On the selected boot device, the bootloader is initiated. This bootloader transfers the Operating System from the selected boot device to the Random Access Memory. Once the OS is transferred, the UEFI gives control over the different components to the OS.

Now that we know how a computer system boots, let's have a little exercise. Open the static site in this task by clicking the green "Static Site" button and follow the instructions to answer the question below.

View Site

Answer the questions below

What is the flag that you received after completing the exercise?

THM{pcSucce55fully5t4rt3d}

✓ Correct Answer

Task 4 Conclusion

We have covered the core components of a computer system and how it boots up. At the moment, you won't realize this, but further on, when learning cyber security concepts, you will often need to recall the function of each of these core components and how they interact with each other. The boot process is a very important concept later on, as it is sometimes targeted by hackers.

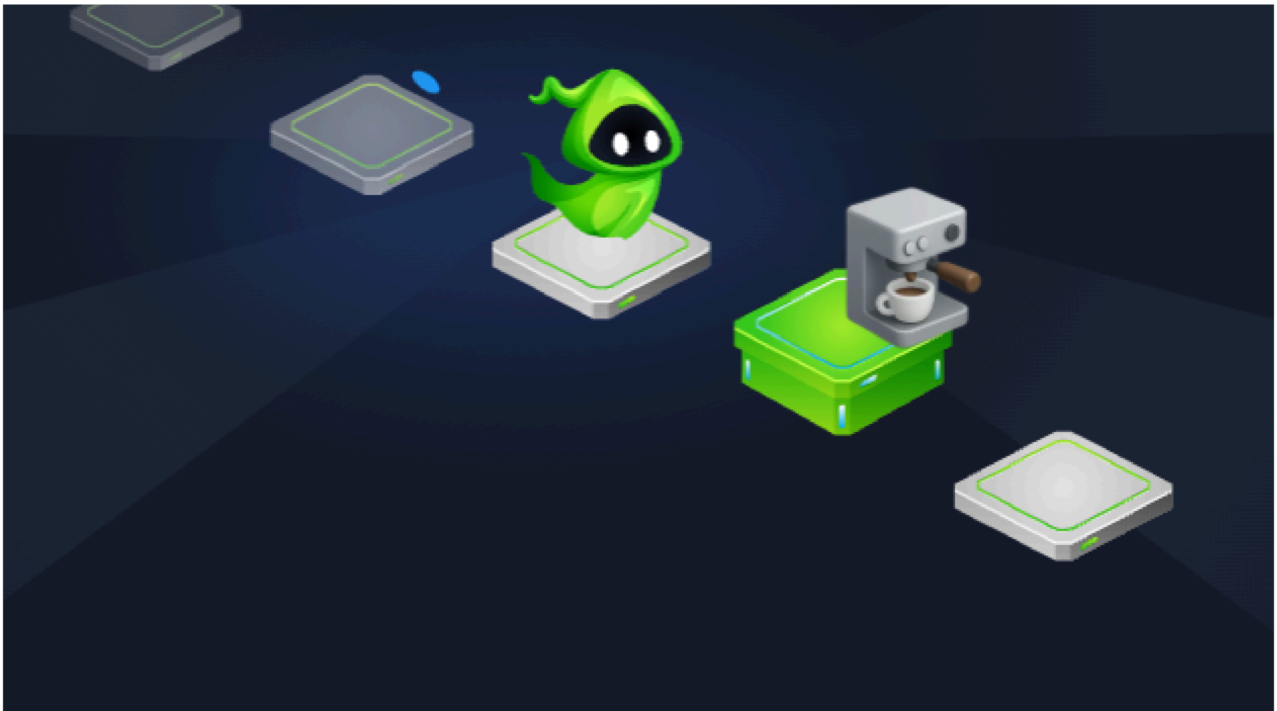
Now that we know the insides of a computer, the next step is to see how the different combinations and specializations of these components lead to diverse types of computer systems. Continue to the next room **Computer Types** (coming soon) to find out.

Answer the questions below

I am ready to discover the different types of computer systems and their function!

No answer needed

✓ Correct Answer



Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks	Points earned	Streak
4	16	2

Computer Types

Premium room

Explore the different types of computers, from laptops to the tiny chips inside your coffee machine.

30 min · 108,846

Save Room · 467 Recommend · Options

Room progress: 14%

Task 1 Introduction

Sophia was trying to connect a new device to her home's WiFi when she noticed something unexpected: "NexusCool Fridge X17." She laughed and wondered if connecting to the `NexusCool Fridge X17` WiFi would let her download a cold meal. Setting the joke aside, she soon realized her neighbor had bought a smart refrigerator, a fridge with a built-in computer capable of connecting to the internet. It made her pause. Computers were no longer just laptops and phones. They were quietly infiltrating everyday objects, from kitchen appliances to doorbells. Sophia had seen strange stories online about smart devices behaving badly, but she had never really stopped to think about what these machines actually were.

She looked forward to learning more about computers during her upcoming summer internship.

Learning Objectives

Upon completion of this room, you will be able to identify and distinguish between different types of computers you use directly, such as laptops and smartphones, and indirectly, such as servers, IoT devices, and embedded systems. You will also understand what makes each type suited to its purpose.

Answer the questions below

Ready to find the hidden computers?

No answer needed

✓ Correct Answer

Task 2 Sophia's Summer of Hidden Computers - Month 1

The Computers You Sit in Front Of

During her first month at Nova Labs, Sophia learned an important lesson.

Sophia learned that...

1. Not all computers are meant to move.
2. Not all computers are meant for people to sit in front of.

She was introduced to four types of computers that often look similar but serve very different purposes.

- Portable everyday computing
- Sustained performance at a fixed location

Let's compare these below:

Computer Type	Screen and Keyboard	Main Purpose
Laptop	Yes	Portable everyday computing.
Desktop	Yes	Sustained performance at a fixed location.
Workstation	Yes	Precision and reliability for professional tasks.
Server	No	Providing services to many users over a network.

How Sophia Learned the Difference

Sophia began with a **laptop**, the computer she already knew best. It was perfect for emails and documents, but when she pushed it with long tasks, it slowed down. Gabriel explained that laptops are built to be portable, and staying cool in a small, battery-powered device is difficult.

Next, she tried a **desktop**. It stayed in one place, used wall power, and had better cooling. The same task ran smoothly for much longer. Desktops are designed for consistency rather than mobility.

When Sophia moved on to professional work, such as simulations and 3D models, Gabriel showed her a **workstation**. It looked like a desktop, but it was built differently. Workstations prioritize accuracy and reliability, using specialized components to reduce errors during long or complex computations.

Finally, Gabriel took her to a room full of machines with no screens at all. These were **servers**. They ran continuously, answering requests from multiple users simultaneously. Sophia never touched them directly, but they powered the tools she used every day.

Answer the questions below

Which computer type usually runs without a dedicated screen and keyboard?

server

✓ Correct Answer

What kind of computer with specialized components would one buy to carry out precision work?

workstation

✓ Correct Answer

Task 3 Sophia's Summer of Hidden Computers – Month 2

By her second month at Nova Labs, Sophia had started noticing computers she had never interacted with directly. The most powerful computer most people own fits in their pocket, but millions more hide inside everyday objects: doors, lamps, coffee machines.

We continue to find computers hiding in everyday objects:

Type	What it is	Examples
Smartphone	Pocket-sized computer optimized for battery life and connectivity	iPhone, Android phone
Tablet	Touch-first computer with larger screen	iPad, drawing tablet
IoT device	Network-connected device with a single purpose	Thermostat, smart doorbell, fitness tracker
Embedded computer	Computer built into another device	Coffee maker controller, automatic door sensor, lamp dimmer chip

IoT vs Embedded: Both can be small and single-purpose. The difference is connectivity. IoT devices connect to a network to report data or receive commands. Embedded computers might not connect to anything; they do their job inside the machine, often for years without anyone knowing they exist.

Sophia walked through automatic doors every day at Nova Labs. She never realized that a tiny computer inside the door frame was detecting her movement and signaling the motor to open. That's embedded computing; invisible, reliable, everywhere.

Answer the questions below

What is the currently most popular pocket-sized computer?

smartphone ✓ Correct Answer

What kind of computer would you expect to find in a coffee machine?

embedded computer ✓ Correct Answer

Computer Types
Sophia's First Day at Nova Labs

Fullscreen Restart Task 80%

Sophia's Fir... | The Hot Lapt... | The Server R... | The Right To... | 5 Graduation

Task 5/5 **Graduation**

5 / 5 Correct

Congratulations! You've mastered computer types!

Your flag is:
THM{8_computer_types}

Nova Labs Certificate

Sophia

has completed the Computer Types Training

"It was great having you at Nova Labs, Sophia. You did great here!"

You've mastered the fundamentals of computer types!

Task 4 Why Computers Come in Different Flavors

Sophia asked the question that had been bothering her all summer:

Sophia asked...

Why not just build one computer that does everything?

"Because every design is a trade-off," replied Gabriel.

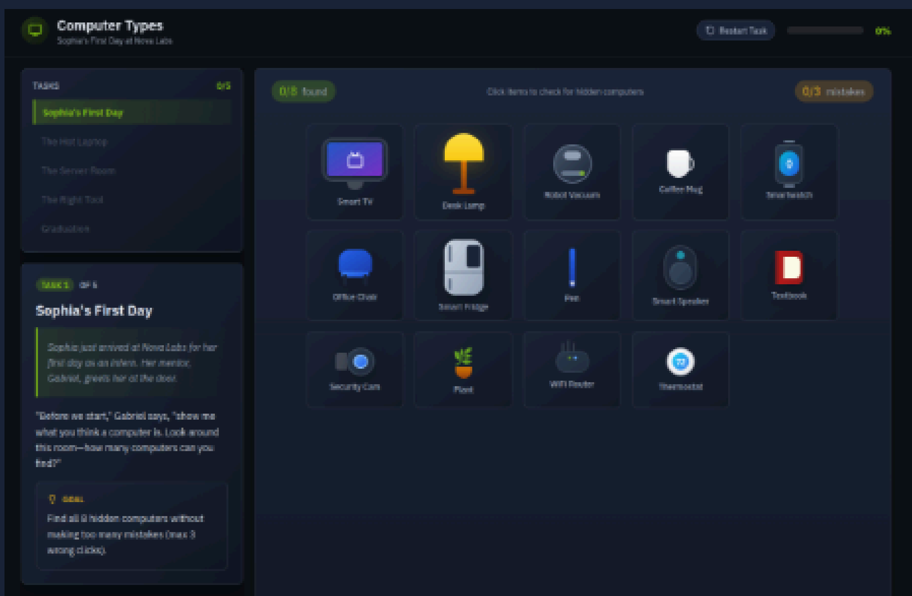
Mobility costs power. Smaller, portable computers must sacrifice sustained performance. Reliability costs money. Servers and critical systems use redundancy, such as extra power supplies and disks, to avoid failure.

Purpose shapes everything. You touch a phone. You ask a server for information. An IoT device works quietly without demanding attention.

There is no best computer. There is only the right tool for the job!

Click the View Site button below, then complete the interactive challenge to get the flag.

View Site



Answer the questions below

Go through the attached static site and get the flag.

THM{8_computer_types}

✓ Correct Answer

Task 5 Summary

Sophia's final report began: "I arrived thinking computers had screens and keyboards. I leave knowing they are everywhere, especially where I do not see them."

In this room, we covered eight types of computers and learned how decisions are made when choosing one over the others. The most critical computers are not always the fastest or flashiest. Sometimes, they are the silent chips that keep doors opening, planes flying, and coffee machines brewing.

Answer the questions below

Room complete!

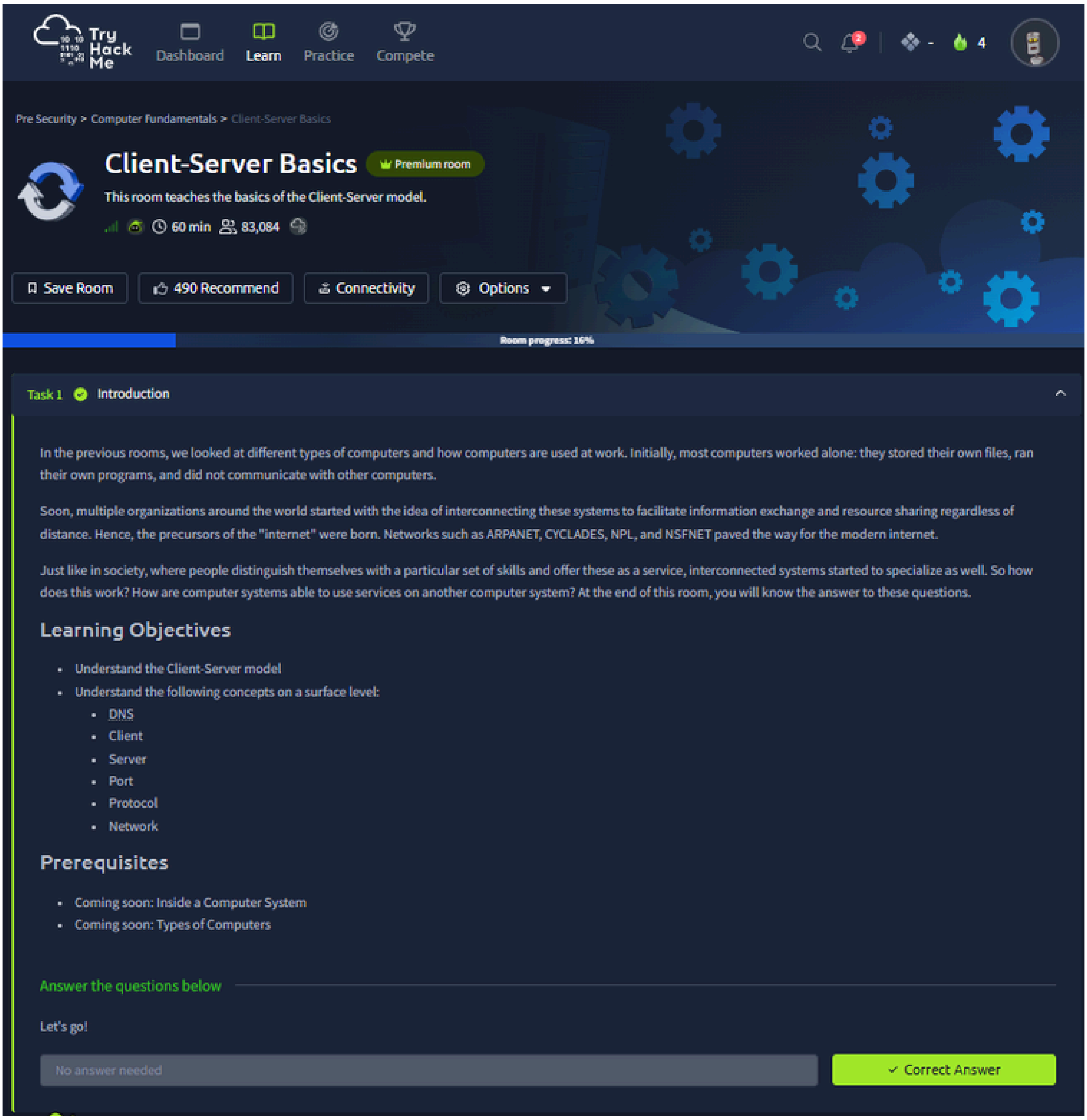
No answer needed

✓ Correct Answer

Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks	Points earned	Streak
5	40	3



Dashboard Learn Practice Compete

Pre Security > Computer Fundamentals > Client-Server Basics

Client-Server Basics

Premium room

This room teaches the basics of the Client-Server model.

60 min 83,084

Save Room | 490 Recommend | Connectivity | Options

Room progress: 16%

Task 1 Introduction

In the previous rooms, we looked at different types of computers and how computers are used at work. Initially, most computers worked alone: they stored their own files, ran their own programs, and did not communicate with other computers.

Soon, multiple organizations around the world started with the idea of interconnecting these systems to facilitate information exchange and resource sharing regardless of distance. Hence, the precursors of the "internet" were born. Networks such as ARPANET, CYCLADES, NPL, and NSFNET paved the way for the modern internet.

Just like in society, where people distinguish themselves with a particular set of skills and offer these as a service, interconnected systems started to specialize as well. So how does this work? How are computer systems able to use services on another computer system? At the end of this room, you will know the answer to these questions.

Learning Objectives

- Understand the Client-Server model
- Understand the following concepts on a surface level:
  - DNS
  - Client
  - Server
  - Port
  - Protocol
  - Network

Prerequisites

- Coming soon: Inside a Computer System
- Coming soon: Types of Computers

Answer the questions below

Let's go!

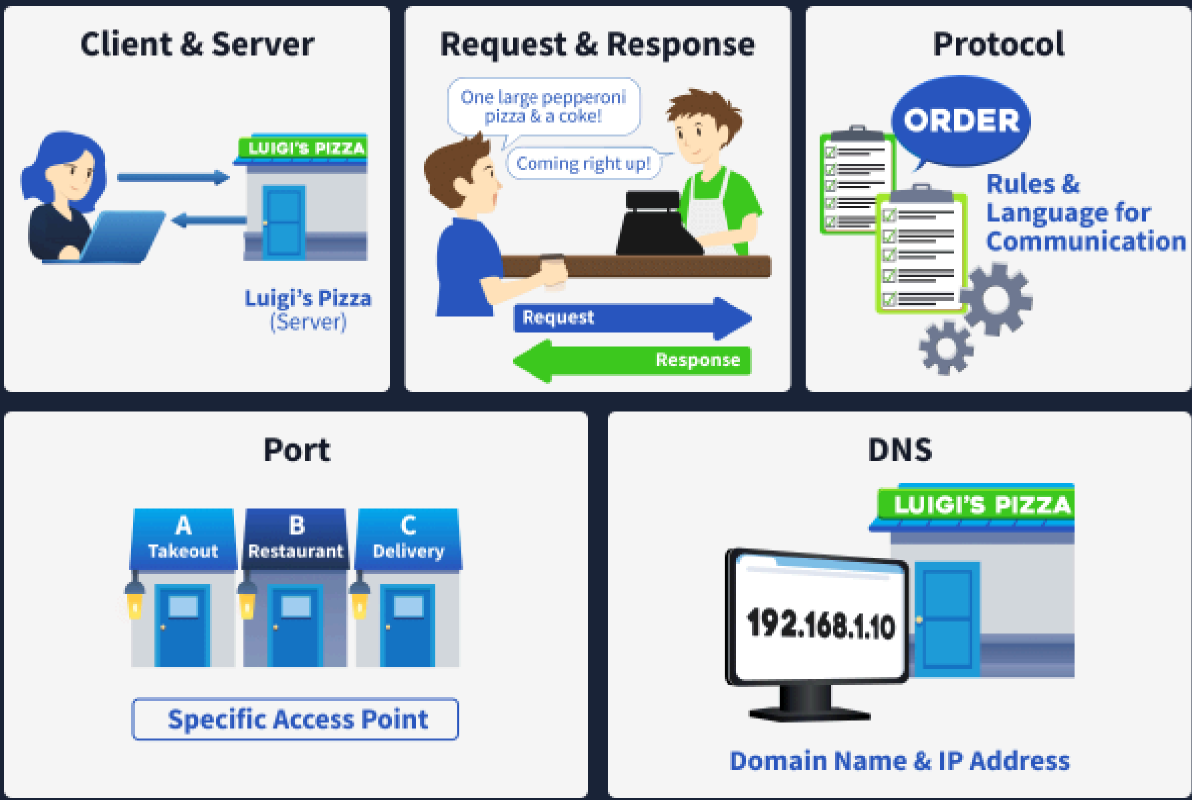
No answer needed | Correct Answer

Task 2 Pizza Delivery

One of the easiest ways to explain how computer systems provide services to other systems is to use an analogy: for example, how shops offer services to their customers. Let's have a look at a pizza takeaway.

It's Friday night, and Alice and Bob feel like having pizza. Alice looks at Luigi's Pizza's menu and tells Bob which pizza she wants. Bob takes the car, enters Luigi's address into his GPS, and starts driving. Once Bob arrives, he enters Luigi's and places his order: "Get me a large pepperoni pizza and a coke." The employee acknowledges the order and starts making the pizza. Once the order is ready, Bob returns home and has a lovely pizza night with Alice.

This seems like a standard, straightforward process. But don't be fooled. Because we are so used to ordering pizza, we have internalized the process. Let's have a closer look at each step and relate it to how computer systems interact.



Service, Client, Server

In our analogy, Bob and Alice choose to use the Pizza takeaway service. Alice is the client who passes her order to Bob, who then passes it to the Server at Luigi's Pizzas. In computer terms, we can translate this as Alice using, for example, a browser to navigate to a website. The browser is the **client** that requests the webpage, and the **server** is the system that serves it.

Note that the client is the one who always initiates the request.

Request and Response

Alice requested a large pepperoni pizza from Luigi's. Be careful, this request was not made to Bob. Bob was the one who brought the request to Luigi's. What is important to realize is that if the request is not formatted correctly or the requested resource is unavailable, we will get an error response. For example, Bob returns to Alice with the message that there were no pepperoni pizzas or that the server did not understand the order.

In computer systems, we can say that Alice used a browser (the client) to request a webpage from a server, which then sent the webpage to the client.

Protocol

One of the things we don't often realize, until we go on a trip to a foreign country, is our language. When Alice formulates her request to Bob, she uses the language Luigi's Pizza understands. Additionally, Alice uses the menu to determine which orders she can place. Bob understood the request (language and order) and went to Luigi's to bring it; he received a response he understood and brought it back to Alice.

We can compare Bob to a protocol that computer systems use to communicate. A protocol defines how a client can communicate with a server. This definition includes:

- Which commands do the client and server understand. E.g., the **get** command.
- How a request is structured. E.g., first the command and then the order.
- What syntax is used. E.g., Alice uses the English language.
- What response should be given to which type of request. E.g., a request for pizza results in receiving the available pizza.
- What response to give to faulty requests. E.g., the server at Luigi's Pizzas says: No pepereonni pizza available

Port

A port is used to identify a specific service running on a system. When a client wants to access a service on a server, it must connect using the correct port.

Imagine that Luigi's takeaway service requires customers to enter through a specific door. Everyone ordering takeaway uses the same door. In the same way, a service on a server listens on a specific port.

Now imagine that Luigi's offers multiple services, such as takeaway, dining in, and delivery. Each service uses a different door: door A for takeaway, door B for the restaurant, and door C for delivery. Similarly, a single server can run multiple services at the same time, with each service identified by a different port.

DNS

When Alice sent Bob her request, only the name of the pizza place was known. With that alone, it is not possible to reach the destination. So, Bob entered Luigi's Pizza into a GPS device, and the device returned the coordinates for Luigi's.

DNS stands for Domain Name Service and works similarly to GPS: when you enter the name of, for example, a website, DNS resolves it to server's location. These location coordinates are called an Internet Protocol (IP) address in computer terms. Imagine this IP as the address to your home (street name, house number, postal code, city, and country), but for computer systems.

Let's look at the next task to see how the client-server model applies when browsing a website.

Answer the questions below

What do we use to identify a specific service on a server?

Port

✓ Correct Answer

What do we call the address of a server?

Internet protocol address

✓ Correct Answer

Task 3 Web Communication in Practice

Hypertext Transfer Protocol (Secure), abbreviated as HTTP(S), is a stateless client-server protocol used for the World Wide Web. This means that each request is processed independently, without the server retaining information about previous requests.

Although the protocol itself is stateless, modern websites and web applications implement mechanisms to introduce statefulness at the application level.
For example, when you log into a website using your credentials, the server creates a session identifier (often stored in a cookie or token) that is sent with each subsequent request.
Without these mechanisms, you would need to authenticate again with every new request, because the server would have no memory of your previous login.

HTTP Commands

In the main specifications that define HTTP (also called Request for Comments, or RFC documents), there are 9 core commands. In HTTP lingo, we use the term method instead of command. Below you can see an overview of these methods:

- GET
- POST
- PUT
- DELETE
- PATCH
- HEAD
- OPTIONS
- CONNECT
- TRACE

We will focus on discussing one of the two most common methods used: GET. We will approach this practically by examining the requests a browser sends when you navigate to a website.

Click on the "Start Lab Machine" button below. This will show a new window in split-screen where we can open a website and inspect a GET request. If the split-screen is not showing, click the blue "Start Split-screen" button at the top of the room.

Your virtual environment has been set up

All machine details can be found at the top of the page.

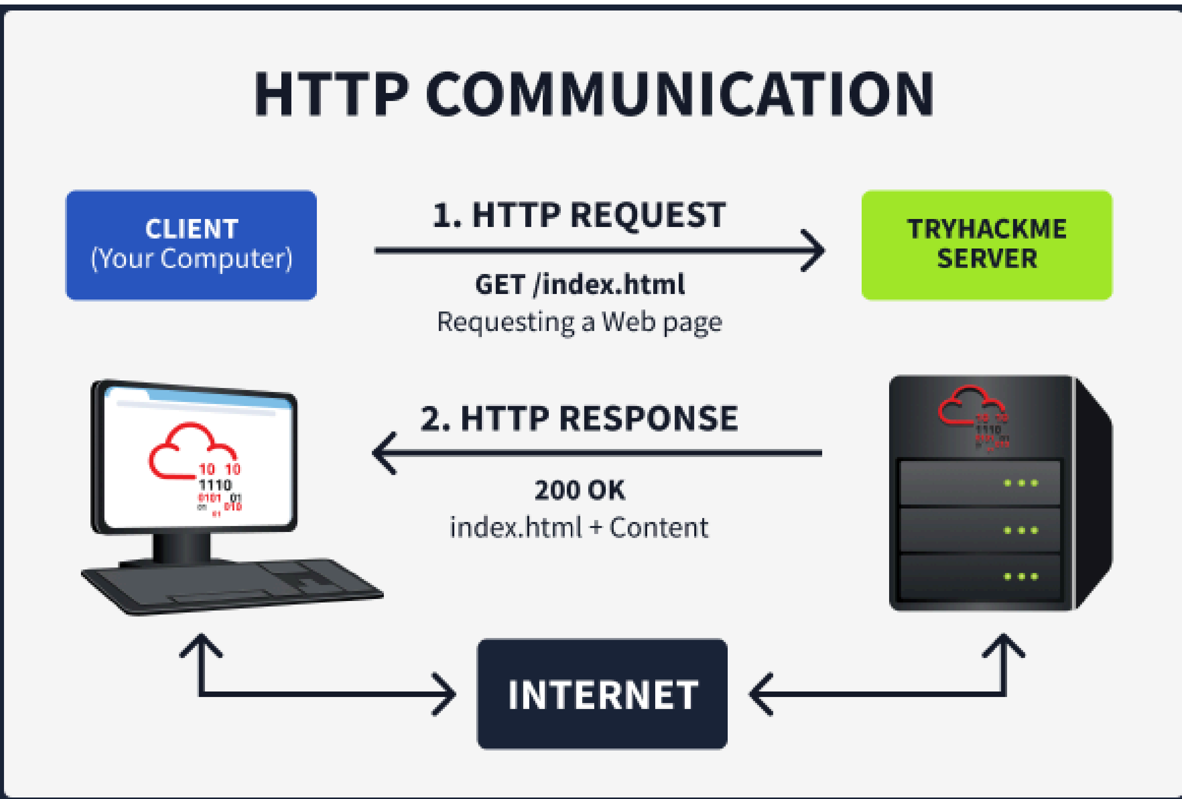
Lab machine

Status: On

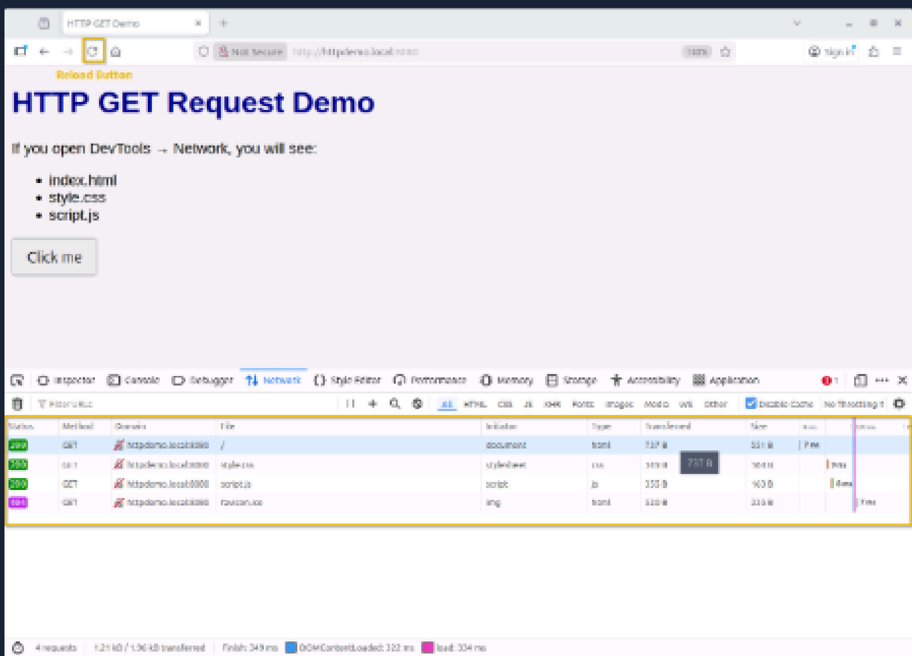
See connection methods

GET

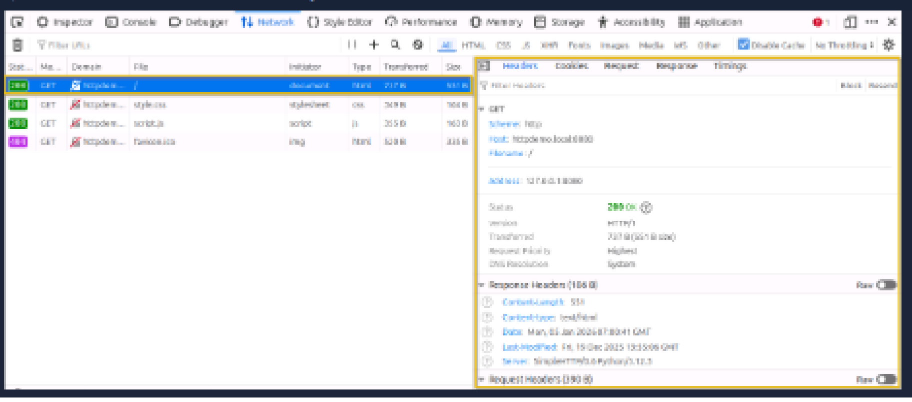
The GET method is actually pretty straightforward. We can use this method to retrieve a resource from web server. For example, `GET https://tryhackme.com/index.php`. This request retrieves the TryHackMe website's homepage. You don't need to type in this request yourself. When you open a browser (this is the client) and type "https://tryhackme.com," the browser constructs the message behind the scenes using information you provide and other fields defined in the HTTP specifications. When the web server receives the request, it sends a response that includes a status code (indicating the type of response) and the requested information. The image below shows the flow of this request.



Let's have a look at an actual GET request and its response. Navigate to the Lab Machine that opened next to this screen, then click the Firefox icon on the Desktop. Once the browser is open, the web page `http://httpdemo.local:8080` should show. Proceed by pressing F12 or right-clicking in the browser window and selecting "Inspect". This will open the Firefox Developer Tools, which allows us to inspect, debug, and analyze web pages and traffic. Click on the "Network" tab as shown in the image below.



Now reload the page by clicking the circular logo highlighted on the image above (next to where you type the URL). You should see multiple GET requests appearing in the Developer Tools window, under the Network tab. Click on the first entry as shown below to see more information.

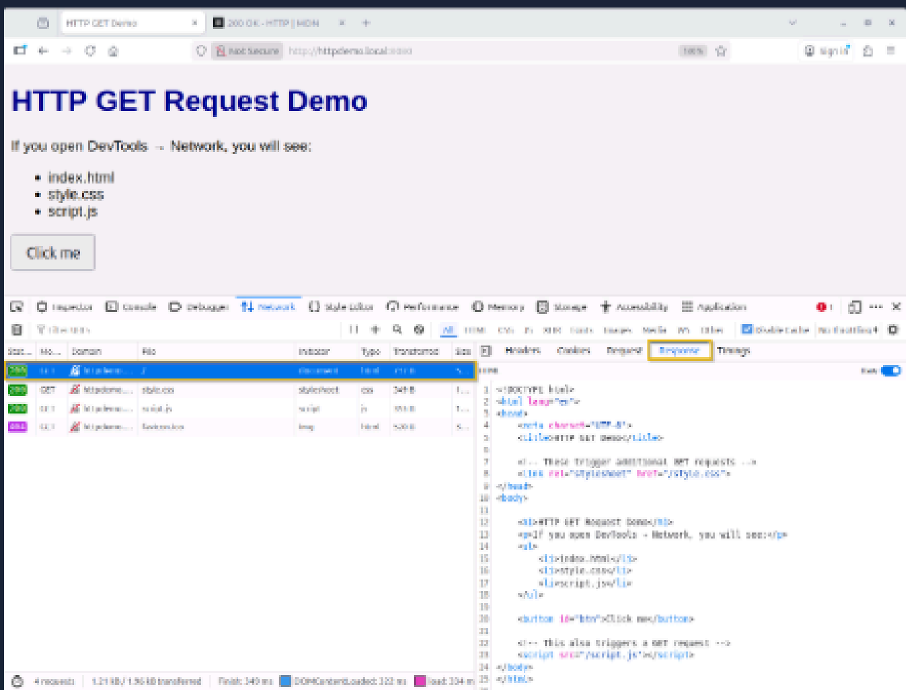


In the right-hand panel, we can see more information about our GET request. We won't go into much detail, but let's have a look at some of these fields.

- **Scheme**: Tells us which protocol was used: HTTP or HTTPS.
- **Host**: Tells us the name of the host we request resources from.
- **Filename**: Indicates which file we requested from the host. In our request, this is "/", which actually translates to "index.html".
- **Address**: Displays the IP address where the website is hosted. In our example, we are hosting the website on the same device. That's why the address 127.0.0.1 is shown.
- **Status**: This field indicates whether the request was successful. In our example, we received a "200 OK" status, which means that the request was successful.

When a request is sent, we will get a response from the server. The response is divided into two parts: the response header and the response body. The response header contains metadata about the response, while the response body contains the requested content.

We can see this response body by clicking on the "Response" tab when displaying the details of a request. In our example, the response contains the index page of the requested website. The image below shows the content in HTML format.



Answer the questions below

What would be the host in the following URL? `https://www.iamlearning.thm/contact`

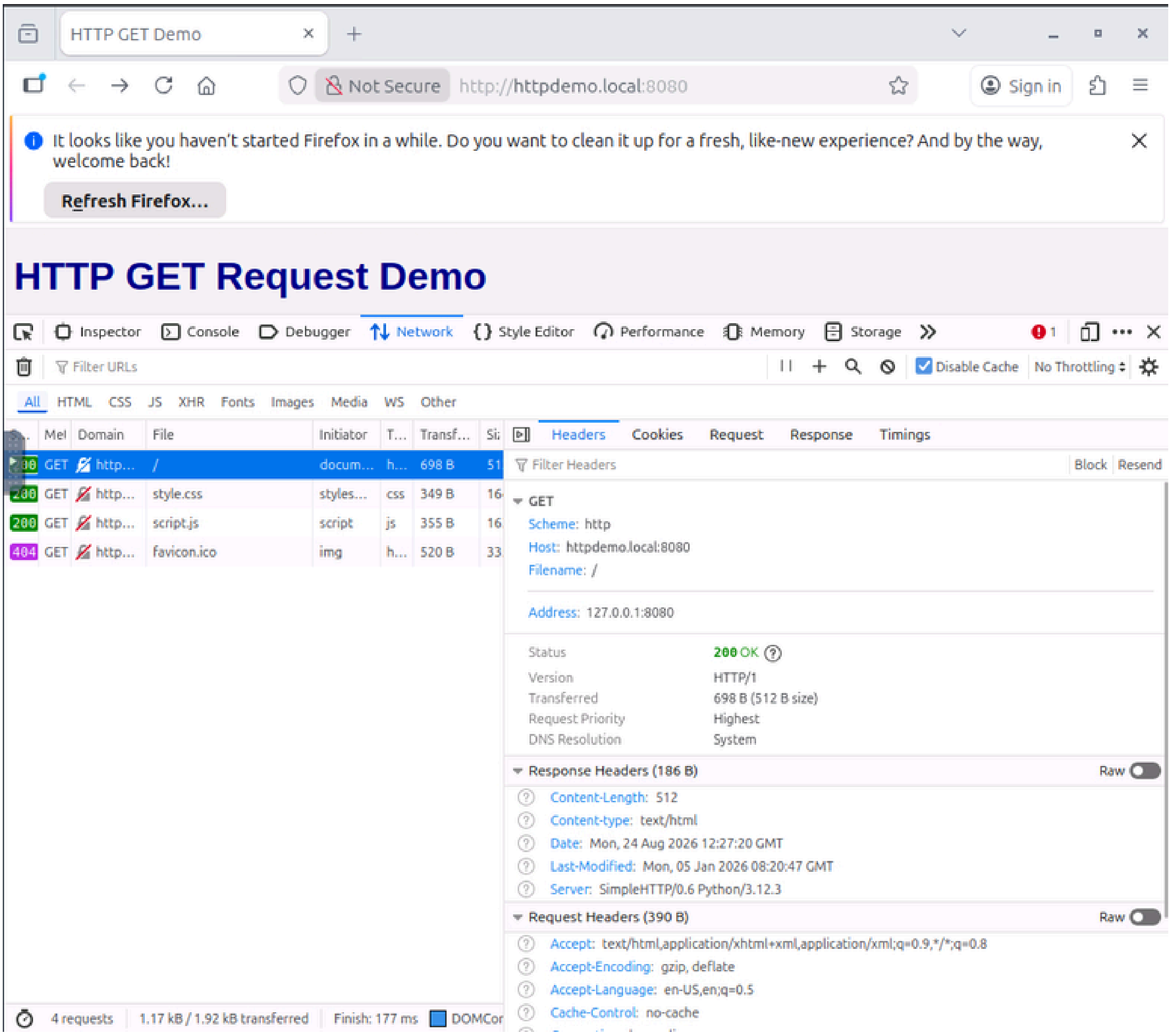
www.iamlearning.thm

✓ Correct Answer

What would be the scheme in the following URL? `https://www.iamlearning.thm/contact`

https

✓ Correct Answer



Task 4 Conclusion

In this room, we have explored how devices on the internet can offer services to each other. We focused on the client-server model, which is similar to ordering a pizza. The client initiates the communication and the server replies.

Then, we continued with an example of the HTTP protocol which is used for websites. We saw a practical example of how a client request and server response actually looks like behind the scenes.

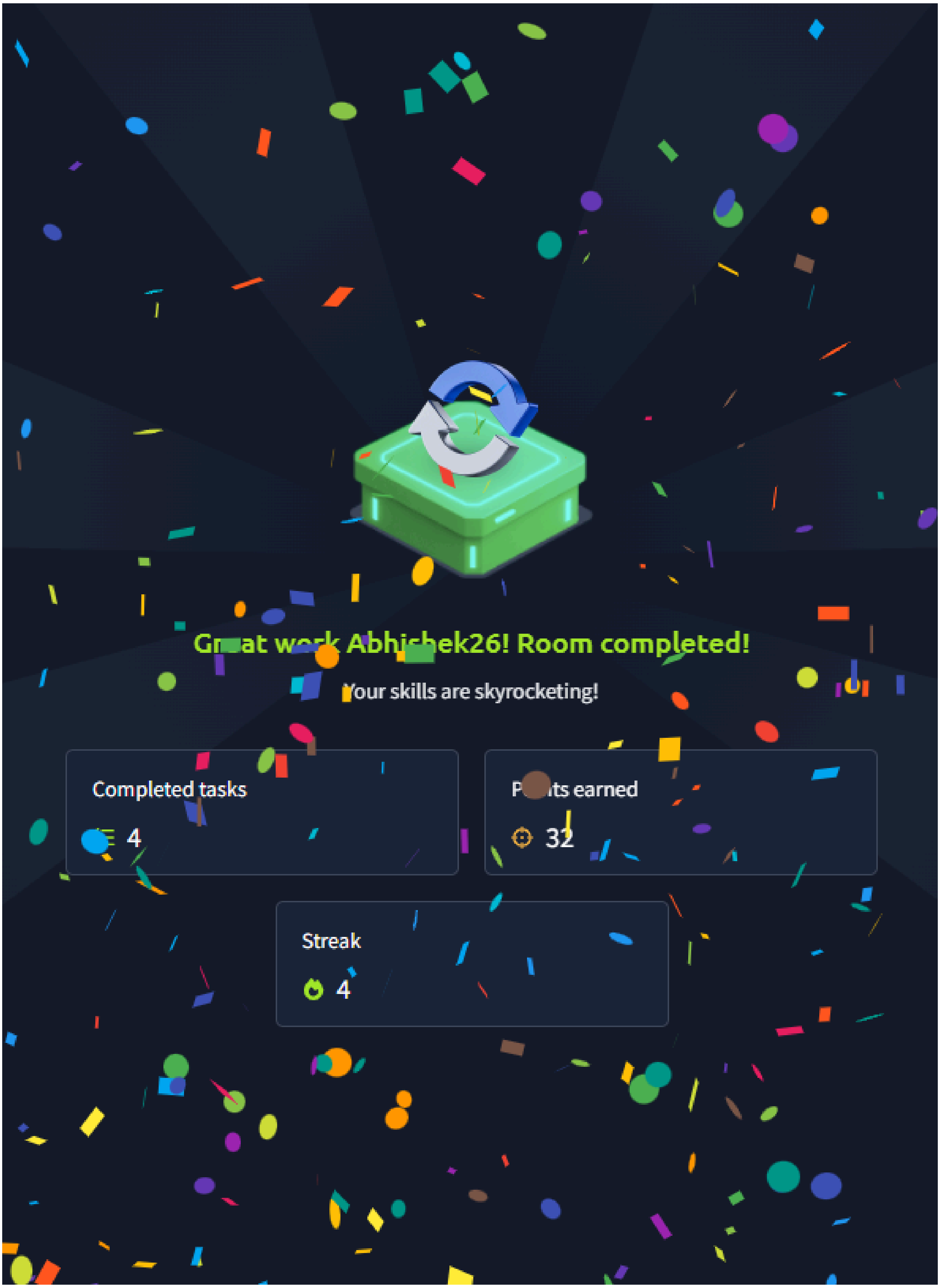
Now that we have seen how services are offered on the internet, let's have a look at the infrastructure that supports them. In the next room we will cover the basics of virtualization.

Answer the questions below

On to the next room!

No answer needed

✓ Correct Answer


Great work Abhishek26! Room completed!
Your skills are skyrocketing!
Completed tasks
4
Points earned
32
Streak
4

Try Hack Me

16 5

Pre Security > Computer Fundamentals > Virtualisation Basics

Virtualisation Basics

Premium room

Learn why virtualisation powers modern IT, improving efficiency and safely isolating environments.

30 min 91,171

Start AttackBox | Save Room | 587 Recommend | Connectivity

Options

Room progress: 10%

Task 1 Introduction

In previous rooms, you learned what the components of a computer are and how they communicate with each other. In this room, you will learn how companies have optimized these computer components to reduce costs and build flexible, scalable systems that power the modern internet through a concept known as **virtualization**.

Have you ever considered how expensive and inefficient it would be if every piece of software or every website required its own physical server?
Virtualization was created to solve exactly this problem.

Your manager asked for help on improving the hardware utilization of a computer that hosts a website. Let's explore the concept of virtualization and help your manager!

Learning Objectives

- Understand why managing applications on individual physical servers is inefficient.
- Learn how virtualization addresses hardware utilization and scalability challenges.
- Understand the components of a lab machine.
- Learn how containers have further optimized hardware utilization for applications.

Prerequisites

Before you start, we recommend that you complete the following rooms:

- Inside a Computer System (coming soon)
- Computer Types (coming soon)
- Client-Server Basics (coming soon)

Answer the questions below

Let's go!

No answer needed

✓ Correct Answer

Task 2 Virtualization Overview

Before the concept of virtualization, the rule of thumb in IT was:
"One server = one application."

In the early days, digital services were run on physical machines, and each machine typically had a single, clear purpose, such as hosting a website or storing data. As businesses added more services, they naturally increased the number of physical servers, and the "one job per box" approach became the standard for building reliable systems.

This meant that if a company wanted to run a website, a database, an email service, and an internal app, they would need separate physical servers for each one. The problems were obvious:

- **High cost**: Buying multiple physical servers is expensive, not just the hardware, but also electricity, cooling, maintenance, and data center space.
- **Low utilization**: Most applications don't use the server's full capacity. Many servers stayed at 5–20% usage, wasting CPU, memory, and storage resources.
- **Slow deployment**: Setting up new physical servers could take days or weeks.
- **Hard to scale**: If an application suddenly needed more resources, you often had to buy yet another server.

In short, companies were paying a lot for hardware that wasn't fully utilized.

The Need for Sharing Hardware Safely and Efficiently

Virtualization introduced a new idea:
“What if multiple applications could share the same physical server safely?”

A virtualization layer, called a **hypervisor**, was introduced to act as a referee between lab machines and allow each virtual computer to behave independently, like a physical computer.

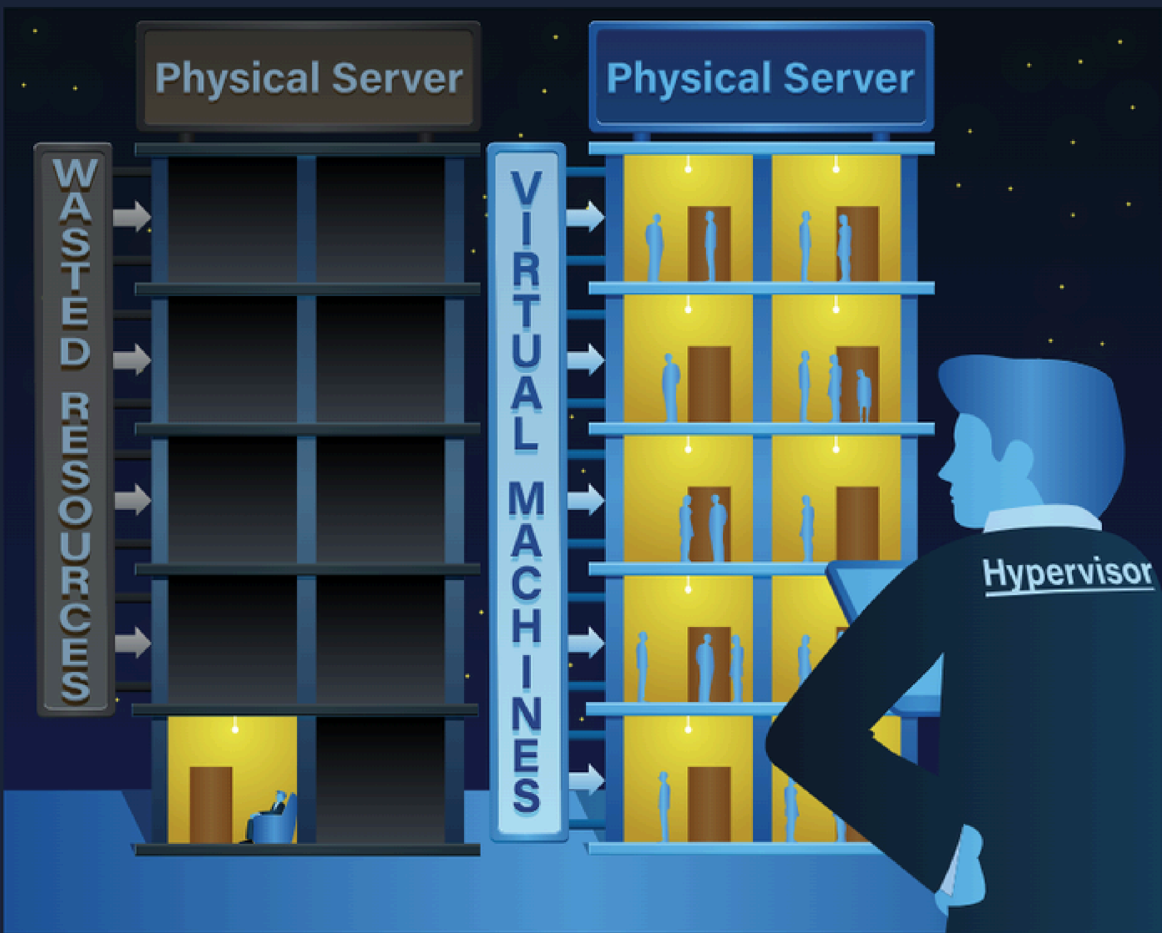
The Building Analogy

Imagine if one single person lives alone in an entire 10-floor building:

- The person uses only one floor but must maintain the entire building: electricity, cleaning, water, and security.
- Most of the building stays empty and wasted.
- It’s expensive, inefficient, and unnecessary for his needs.

Now imagine dividing the building into **separate apartments**:

- Each apartment has its own door, walls, kitchen, and privacy.
- Different people can live independently without bothering each other.
- They all share the building’s main structure: electricity, water, and elevators, making it **cheaper and more efficient** for everyone.



This is virtualization:

- The building = the physical server
- The apartments = lab machines
- The tenants = applications or operating systems
- The building manager = the hypervisor (the software that divides the building safely)

Each virtual computer, known as a Lab Machine (VM), **acts as an independent system** with its own operating system, apps, and settings, even though they all share the same physical hardware underneath.

Answer the questions below

What does virtualization enable multiple applications to share?

physical server

✓ Correct Answer

What is the name of the software that manages the resources for each lab machine?

hypervisor

✓ Correct Answer

Task 3 Virtualization Components

Hypervisor (The Building Manager)

A **hypervisor** is the core technology behind virtualization. It's the software that creates and manages lab machines.

It is a special piece of software that:

- Divides a physical computer into multiple virtual ones.
- Gives each lab machine its own share of CPU, memory, and storage.
- Keeps everything isolated and safe.
- Manages the lifecycle of lab machines (start, stop, pause, clone, delete).

Hypervisors have two main types of implementation, each of which is used for specific scenarios, from home labs to large data centers:

- **Type 1** hypervisors run directly on the physical hardware, making them fast, efficient, and ideal for servers and professional environments.
- **Type 2** hypervisors run within an existing operating system, making them easier to install and ideal for learning, testing, or small setups.

Below is a table showing which use case is best suited to each hypervisor type. The use cases can run on both hypervisor types, but it's not the best approach given the main objectives of each.

Use Case	Type 1	Type 2
Test Malicious Files		X
Production Server	X	
Database Server	X	
Software Testing		X
Kali Linux		X
Data Center	X	

When using virtualization to test malicious files, care should be taken to ensure that the host machine does not become infected by the malware being tested in the guest machine. One approach is to use different operating systems for the guest and host machines, or to isolate the guest machine so that it does not communicate with the host.

Lab Machines (The Apartments)

A **Lab Machine (VM)** is a virtual computer created by the hypervisor.
Even though it's virtual, it behaves as a real machine:

- It has its own virtual CPU, RAM, storage, and network.
- It can run any operating system (Windows, Linux, etc.).
- It's completely isolated from other VMs. This means that if one VM breaks, the others continue to work.

You can deploy VMs on your own computer using tools such as Oracle VirtualBox and VMware Workstation. This type of software acts as a type 2 hypervisor and lets you run multiple operating systems, such as Windows, Linux, and macOS.

Since you have learned what a hypervisor and VM are, let's take some examples where you might need them:

- You need to work on a different OS like Kali Linux, but you can't buy another whole system, so you install a hypervisor and run a Kali Linux VM on it.
- You want to test whether a file is malicious, so you set up an isolated lab machine to protect your main computer from being infected.

Containers (The Rooms Inside the Apartment)

A container is a lightweight, isolated environment that runs a single application and all the necessary components to support it. Instead of bringing a whole separate operating system, a container borrows the core of the existing system by running on the kernel, which is the part of an operating system that communicates with the hardware and manages resources such as memory and running programs.

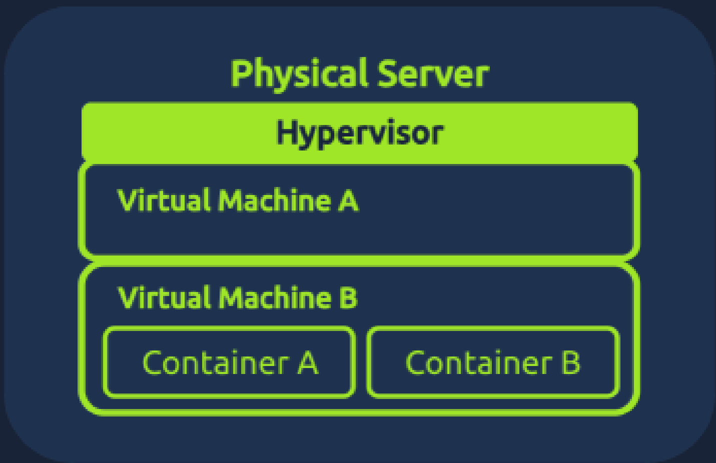
Because containers share this kernel, they start quickly and use fewer resources than full lab machines, but it also means they must match the host system's type. For example, you can't run a Windows container on a Linux machine.

Containers behave like small, self-contained spaces because:

- They package the application and its dependencies (libraries, tools, versions).
- They share the host's operating system, so they start almost instantly.
- They remain isolated from each other, so a misbehaving container doesn't affect the others.
- They can run consistently on any machine, making them perfect for development, testing, and scalable deployments.

The easiest way to deploy containers in a VM is using Docker.
Docker is an open-source software platform that simplifies the process of building, deploying, and running applications using containerization.

The image below illustrates the relationship between Hypervisors, VMs, and containers:



In summary, VMs provide the "full apartment" with maximum separation and flexibility, while containers offer lightweight "rooms" ideal for scalable, fast-deploying applications.

Answer the questions below

Suppose a user wants to deploy a study lab on their machine to practice some exercises for a cyber security certification. Which type of hypervisor will they use?

type 2 ✓ Correct Answer

Suppose a company wants to host multiple small applications in the same lab machine. What should they use?

containers ✓ Correct Answer

Task 4 Managing Virtual Machines

After learning about virtualization, you were hired to be responsible for the virtual environment for AutoGalo, a company that has recently started using lab machines.
In this company, they use an application called Virtualization Manager. This application provides a clear view of the overall status of an entire virtualization environment, providing details of each virtualized instance and the physical hosts. It also enables you to run actions and manage the VMs you create.

Your manager has asked you to investigate an issue with the email service. Essentially, everyone in the company stopped receiving emails today, and no one knows what happened.

Open the Virtualization Manager site by clicking the `View Site` button below, and let's investigate!

View Site

Analyzing Lab Machine States

In the home screen, you should see three main sections:

- **Summary:** Provides a generic view of the state of the environment.
- **Lab Machines:** Provides details of each VM and enables you to run actions on them.
- **Hosts:** Displays usage and performance details for each physical server.

Go to the Lab Machines section and look for the VM with the name `Mail-SERVER`:

Virtual Machines

Name	Status	Actions	CPU	Memory	Disk	IP Address	Uptime	Host
WebServer-PROD	Running		4 cores	8 GB	100 GB	192.168.1.10	2 days, 5 hours	HV-PROD-01
AppServer-STAGE	Stopped		2 cores	4 GB	50 GB	192.168.1.12	N/A	HV-PROD-01
DB-Cluster-01	Running		8 cores	32 GB	500 GB	192.168.1.11	1 week, 3 days	HV-PROD-02
Backup-SERVER	Running		2 cores	16 GB	1000 GB	192.168.1.13	4 days, 12 hours	HV-PROD-02
Dev-Environment	[illegible]		4 cores	16 GB	200 GB	192.168.1.14	N/A	HV-PROD-02
Monitoring-SYS	Running		2 cores	8 GB	100 GB	192.168.1.15	3 weeks, 2 days	HV-PROD-02
Mail-SERVER	Error		2 cores	8 GB	250 GB	192.168.1.16	3 hours	HV-PROD-02
FileServer-01	Running		4 cores	16 GB	500 GB	192.168.1.17	2 weeks, 6 days	HV-PROD-02
File-Server-2	Running		4 cores	16 GB	500 GB	192.168.1.18	1 week, 4 days	HV-PROD-02
File-Server-3	Running		4 cores	16 GB	500 GB	192.168.1.19	6 days, 1 hour	HV-PROD-02

Looks like this lab machine has entered an `Error` state. Let's try rerunning this VM using the `Blue Square Button` to restart it.

Virtual Machines

Name	Status	Actions	CPU	Memory	Disk	IP Address	Uptime	Host
WebServer-PROD	Running		4 cores	8 GB	100 GB	192.168.1.10	2 days, 5 hours	HV-PROD-01
AppServer-STAGE	Stopped		2 cores	4 GB	50 GB	192.168.1.12	N/A	HV-PROD-01
DB-Cluster-01	Running		8 cores	32 GB	500 GB	192.168.1.11	1 week, 3 days	HV-PROD-02
Backup-SERVER	Running		2 cores	16 GB	1000 GB	192.168.1.13	4 days, 12 hours	HV-PROD-02
Dev-Environment	Stopped		[illegible]	16 GB	200 GB	192.168.1.14	N/A	HV-PROD-02
Monitoring-SYS	Running		2 cores	8 GB	100 GB	192.168.1.15	3 weeks, 2 days	HV-PROD-02
Mail-SERVER	Error		2 cores	8 GB	250 GB	192.168.1.16	3 hours	HV-PROD-02
FileServer-01	Running		4 cores	16 GB	500 GB	192.168.1.17	2 weeks, 6 days	HV-PROD-02
File-Server-2	Running		4 cores	16 GB	500 GB	192.168.1.18	1 week, 4 days	HV-PROD-02
File-Server-3	Running		4 cores	16 GB	500 GB	192.168.1.19	6 days, 1 hour	HV-PROD-02

After running again, it appears to be working correctly, with no errors!

Creating a Lab Machine

As part of your routine, you create lab machines for other teams. After resolving the email server issue, you received a task to create a VM to support the marketing team's website.

Now, let's create a VM to host their marketing website. In the Lab Machines section, locate the `+ Create VM` button on the top right side of the Lab Machines section and click on it.

You should fill out the form with how much hardware your lab machine will be using:

- Name: `Marketing-VM`
- CPU Cores: `4`
- Memory (GB): `8`
- Disk Size (GB): `100`

Then click on `Create VM`

Now, at the top of the list of Lab Machines, you should see your created VM:

Analyzing Hardware Usage

Another routine task you have is to manage the health of the physical servers and report status to your manager. Go to the Hosts section at the bottom of the page:

By analyzing it, we can identify:

- `HV-PROD-01` has the capacity to host more VMs.
- `HV-PROD-02` is almost operating at 100% of its capacity, and we might need to report this!
- `HV-BACKUP-01` is disconnected and does not host any VMs.

Now, answer the following task questions to conclude your shift as the responsible for virtualization in AutoGalo!

Answer the questions below

What is the name of the lab machine that has been running for the longest time?

Monitoring-SYS

✓ Correct Answer

What is the name of the lab machine that is using the biggest amount of memory?

DB-Cluster-01

✓ Correct Answer

How many VMs are in the running state after you solved the issue on `Mail-SERVER`?

8

✓ Correct Answer

What is the name of the physical machine that is hosting most of the VMs?

HV-PROD-02

✓ Correct Answer

Task 5 Conclusion

In this room, you learned why **virtualization** is such a critical foundation in modern IT, both for maximizing hardware efficiency and for safely isolating environments.

Key Terminology

Let's quickly review some concepts we learned in this room:

- **Virtualization:** Enables a single physical computer to act like multiple separate computers.
- **Hypervisor:** The "manager" software that makes and runs the virtual computers.
- **Lab Machine (VM):** A whole virtual computer inside the real one, with its own system.
- **Container:** A small, isolated box for one app that shares the same system as the host.
- **Container Images:** A pre-packed recipe/template used to create containers.
- **Network Ports:** Special numbered entry points that apps use to talk over the network.

We also concluded that the key benefits of virtualization are:

- Cost savings
- Better resource usage
- Safe testing for cyber security
- Faster deployment
- Flexibility
- Portability
- Scalability
- Centralized Management

Further Learning

Through hands-on practice, you experienced how **virtualization** simplifies and accelerates application deployment, providing a fast and secure way to run applications consistently.

With these fundamentals, you are ready to explore the Cloud concept in the Cloud Computing Fundamentals room, which uses virtualization, containerization, and automation to provide scalable, on-demand services!

Answer the questions below

I'm ready to Cloud!

No answer needed

✓ Correct Answer

Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks	Points earned	Streak
5	64	5

Try Hack Me   Dashboard   Learn   Practice   Compete   17   6

Pre Security > Computer Fundamentals > Cloud Computing Fundamentals

Cloud Computing Fundamentals

Premium room

Discover how cloud computing helps businesses move faster, do more, and scale with less effort.

30 min   86,457

Start AttackBox | Save Room | 496 Recommend | Connectivity | Options

Room progress: 11%

Task 1 Introduction

Imagine you have a fantastic idea for an app that helps students practice cyber security, and you host it on your own computer in your country. But what can you do when users from other parts of the world try to access it and experience lag? What if many students connect at the same time, or your computer is turned off? These limits make it hard for the app to grow. **That's when cloud computing comes to play and solves these problems!**

The cloud is built on top of technologies you already learned, like virtualization and containers. These enable running many applications efficiently on shared infrastructure and quickly creating or changing environments when needed.

In this room, we will explore the basics of cloud computing and how this impacts the way modern applications are built, deployed, and used every day.

Learning Objectives

- What is cloud computing
- Service models of cloud (IaaS, PaaS, SaaS)
- Cloud Types (Private/Public/Hybrid)
- Benefits of cloud computing
- How big companies are using the cloud

Prerequisites

- Virtualisation Basics room

Answer the questions below

Let's go!

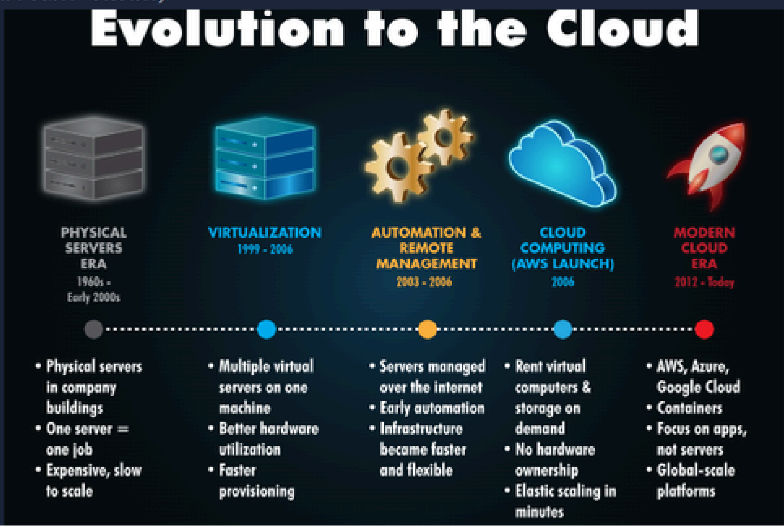
No answer needed   ✓ Correct Answer

Task 2 Cloud Computing Overview

Cloud computing is the perfect solution for the challenges in the application you've designed, such as moving your files from a single laptop to online storage that you can access anywhere. Instead of running your app on one computer in one country, the cloud lets you use computing resources over the internet. This makes your application easier to access, more reliable, and ready to grow as more students start using it.

How Servers Evolved to Cloud

Before we start diving into cloud details, it's helpful to understand that cloud computing did not appear suddenly. It is the result of many years of changes in how servers were used and managed. At each step, businesses looked for ways to reduce costs, use resources more efficiently, and make their applications easier to run and scale. The timeline below shows this evolution, from physical servers to the cloud we use today:



Cloud Benefits and Characteristics

After seeing how applications evolved from physical servers to the cloud, it's becoming clear why cloud computing is so widely used today. The cloud was designed to address common problems, including limited capacity, high costs, and slow growth.
The following benefits and characteristics explain how cloud computing makes applications easier to run, scale, and manage:

- **Scalability:** Easily scale up or down as your application's needs change.
- **On-demand self-service:** Create or remove servers and storage instantly, without waiting for hardware.
- **Pay only for what you use:** You are charged based on usage, not upfront costs.
- **Security:** Cloud providers protect the infrastructure with strong security measures.
- **High availability:** Applications keep running even if part of the system fails.
- **Global access:** Your application can be accessed by users anywhere in the world.

In simple terms, the cloud enables IT resources to be flexible, cost-effective, and easier to manage.

Types of Cloud

The flexibility provided by cloud computing allows applications to be run in different ways, depending on your needs and level of control. Because of this, cloud providers offer multiple models for deploying and using applications, each suited to different scenarios.

Let's start with the **deployment types** you can choose for a cloud environment:

- **Public Cloud:** Used by startups, websites, and global apps because it is affordable, easy to scale, and requires no infrastructure management. Public cloud services are preferable for nearly every use case.
- **Private Cloud:** Used by banks, healthcare, and government organizations because it offers greater control, customization, and compliance for sensitive data.
- **Hybrid Cloud:** Used by companies like e-commerce platforms that need to keep sensitive data private while still scaling publicly during high demand.

Just like there are different ways to deploy a cloud environment, there are also **different ways to use cloud services**. Depending on your experience and needs, you can choose the level of responsibility that fits your application.

Let's look at the main cloud service models:

- **Infrastructure as a Service (IaaS):** You rent basic computing resources such as virtual servers, storage, and networking. You are responsible for managing the operating system and your application, while the provider manages the physical hardware.
- **Platform as a Service (PaaS):** The cloud provider manages the infrastructure and the operating system. You focus on building, deploying, and running your application without worrying about servers.
- **Software as a Service (SaaS):** You use a complete application over the internet. The provider manages everything, and you access the software through a browser or app, for example, Gmail or Zoom.

Think of cloud service models like different ways of renting a place to live:

Major Cloud Vendors

There are several cloud vendors offering a variety of services, but Amazon Web Services (AWS) is the industry leader, with the most extensive offerings and global reach. Other well-known cloud providers include:

- **Microsoft Azure:** A strong competitor, especially in enterprise and hybrid cloud environments.
- **Google Cloud Platform (GCP):** Known for powerful data analytics, AI, and machine learning tools.
- **Alibaba Cloud:** A major player in Asia, offering competitive cloud services globally.
- **IBM Cloud:** Focuses on hybrid cloud and AI-driven solutions for businesses.
- **Oracle Cloud:** Focuses on enterprise applications and databases.

Each of these vendors offers a range of services, but AWS remains the most popular due to its vast infrastructure and support for businesses of all sizes.

How Companies Are Using the Cloud

- **Netflix** runs its entire platform on AWS so it can scale globally, stay online during peak demand, and stream content reliably to millions of users at once.
- **Spotify** uses the cloud to handle millions of songs and users, scaling quickly when new music or features are released.
- **Instagram** relies on the cloud to store massive amounts of photos and videos and deliver them fast to users around the world.
- **Online stores** use the cloud to handle traffic spikes during black friday without buying permanent infrastructure.

These companies use the cloud because it lets them scale easily, reduce costs, stay reliable, and **focus on improving their products instead of managing hardware**.

Next, you'll apply these same ideas by deploying your cyber security training app in a simulated cloud environment!

Answer the questions below

What is the characteristic of cloud environments that enables you to handle an unexpected increase in access to your application?

Scalability

✓ Correct Answer

What is the most common type of cloud deployment used?

Public Cloud

✓ Correct Answer

Suppose you want to deploy an application to the internet, focusing only on application development and leaving infrastructure to others. What type of cloud service is the best?

PaaS

✓ Correct Answer

Task 3 Deploying a Cloud Instance

So far, you've learned what cloud computing is and why companies use it. Now it's time to see those ideas in action by deploying a cloud environment to launch your cyber security app training!
In this exercise, you'll use a cloud interface similar to the AWS platform. The goal is not to memorize buttons, but to understand **how cloud resources are easily created and managed** in a real-world scenario.

Open the Cloud Console site by clicking the `View Site` button below, and let's create your cloud environment!

View Site

Basic Cloud Terminology

To complete this exercise, you only need to understand a few basic concepts from AWS:

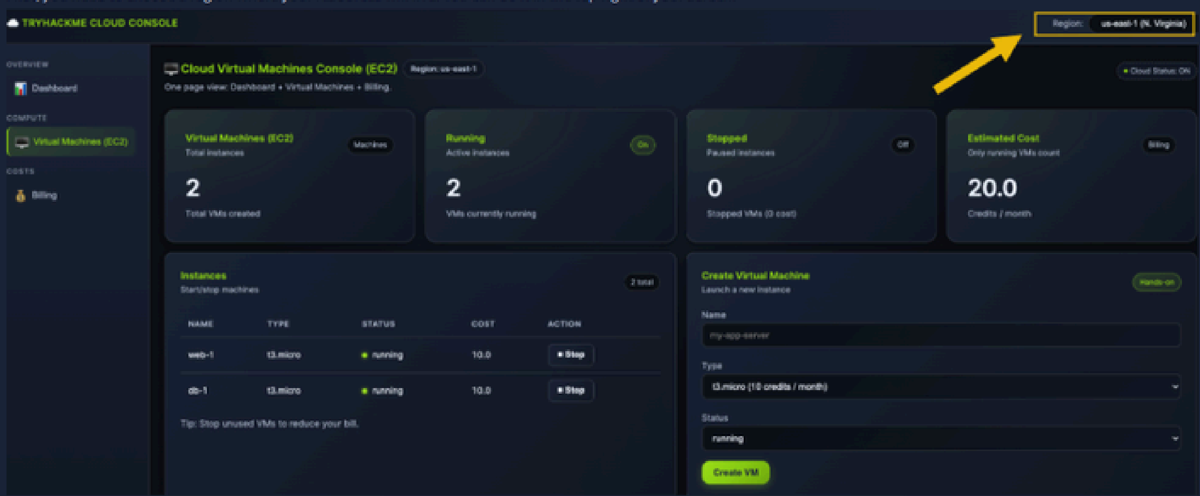
- **EC2 (Virtual Computer / Server):** EC2 represents a virtual computer in the cloud. Just like a real computer, it has a CPU and memory (RAM) and can run applications. Whenever you add an EC2 instance, you are adding a computer to your environment.
- **Instance Type (for example: t2, t3, m5):** Instance types describe how powerful the virtual computer is. Some have more CPU and RAM and are therefore more expensive. You choose the Instance Type based on your needs, knowing that:
  - Bigger instances = more power + higher cost
  - Minor instances = less power + lower cost

Deploying Your Environment

You will create three virtual computers (EC2 instances) to host your cyber security training application. This aligns with the Infrastructure as a Service (IaaS) model you previously learned, as cyber security practices often require full access to the operating system. This allows you to install tools, configure the system, and safely simulate attacks and defenses, just like in real-world environments!

Picking a Region

First, you need to choose a region where your resources will live. You can do it in the top right of your screen:

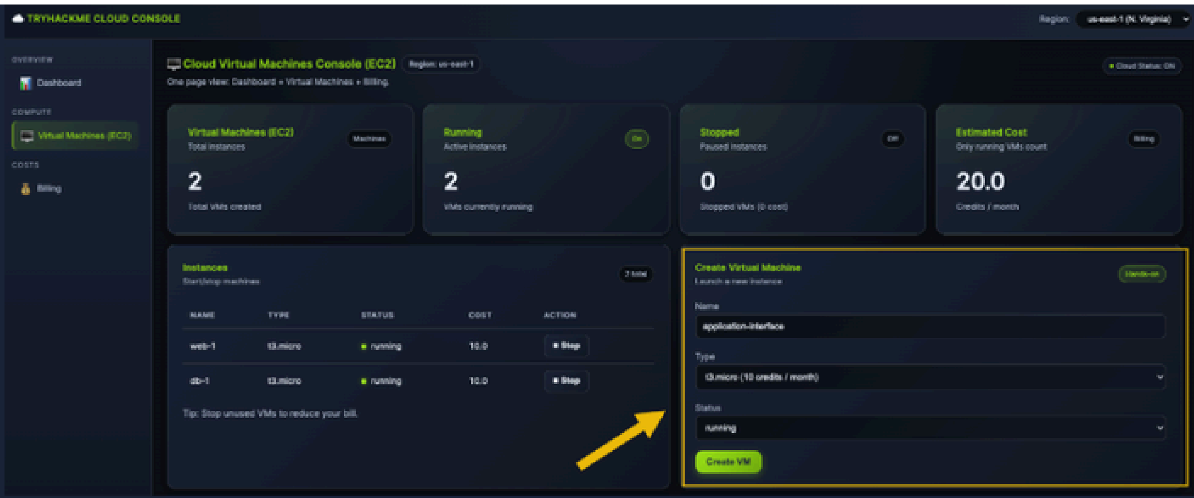


A region represents a geographical location, for example, Europe or North America.

Creating Lab Machines

Now, go to the `Create Lab Machine` block on the right side of the page to create the lab machines for your application.
First, let's create your application interface machine. Set the following configuration:

- Instance Name: `application-interface`
- Instance Type: `t3.micro`
- Status: `running`



Now, let's create two testing computers for users to practice their cyber security skills.
Since these are test machines, let's use a more powerful instance type: `m5.large`.

Machine 1:

- Instance Name: `study-machine-1`
- Instance Type: `m5.large`
- Status: `running`

Machine 2:

- Instance Name: `study-machine-2`
- Instance Type: `m5.large`
- Status: `running`

After creating the two new machines, your `Instances` block should look like this:

Instances
Start/stop machines
5 total

NAME	TYPE	STATUS	COST	ACTION
web-1	t3.micro	running	10.0	Stop
db-1	t3.micro	running	10.0	Stop
application-interface	t3.micro	running	10.0	Stop
study-machine1	m5.large	running	70.0	Stop
study-machine2	m5.large	running	70.0	Stop

Tip: Stop unused VMs to reduce your bill.

Billing Analysis

Let's analyze how much credit is costing our environment by navigating to the `Billing` section at the bottom of the page.
There, you can check how much each type of instance is costing you, as well as the total.

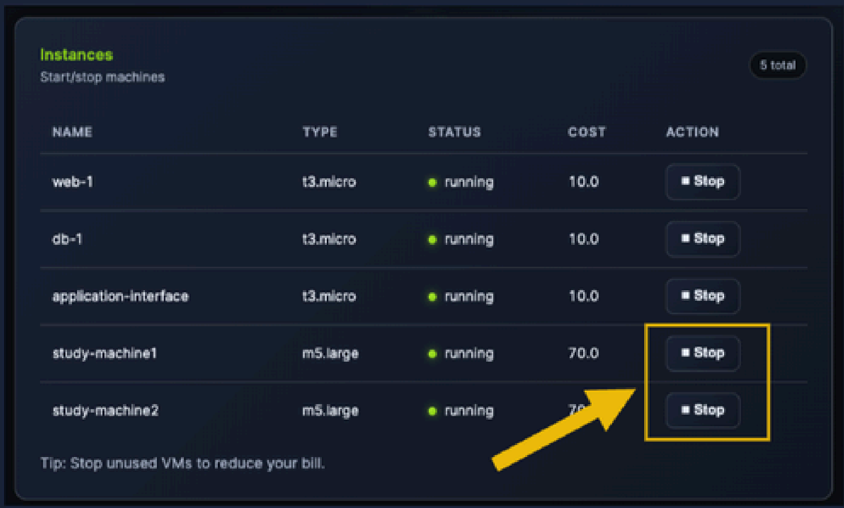
Billing
Only running VMs add cost
Summary

NAME	TYPE	STATUS	MONTHLY COST
web-1	t3.micro	running	10.0
db-1	t3.micro	running	10.0
application-interface	t3.micro	running	10.0
study-machine1	m5.large	running	70.0
study-machine2	m5.large	running	70.0

Total estimated: 170.0 credits / month

…ntly, you are still developing your application, so users have not yet accessed your platform. We can optimize costs by stopping the two study machines and reviewing the …ost of your environment.

Go to the `Instances` block and click the `Stop` button for both `study-machine-1` and `study-machine-2`.



Instances
Start/stop machines
5 total

NAME	TYPE	STATUS	COST	ACTION
web-1	t3.micro	running	10.0	Stop
db-1	t3.micro	running	10.0	Stop
application-interface	t3.micro	running	10.0	Stop
study-machine1	m5.large	running	70.0	Stop
study-machine2	m5.large	running	70	Stop

Tip: Stop unused VMs to reduce your bill.

Go back to the `Billing` section, and you should see that your cost was reduced considerably:

Billing
Only running VMs add cost
Summary

NAME	TYPE	STATUS	MONTHLY COST
web-1	t3.micro	running	10.0
db-1	t3.micro	running	10.0
application-interface	t3.micro	running	10.0
study-machine1	m5.large	stopped	0.0
study-machine2	m5.large	stopped	0.0

Total estimated: 30.0 credits / month

This exercise is a clear example of how easily you can change and manage your costs in a cloud environment.

Answer the following questions to conclude your learning about cloud computing!

Answer the questions below

What is the total cost of credits of the entire environment if `study-machine-1` and `study-machine-2` are stopped?

30

✓ Correct Answer

How many credits does an `m5.large` EC2 instance cost per month?

70

✓ Correct Answer

What is the total cost of credits if **only the new instances** we created are running?

150

✓ Correct Answer

What would be the total running cost of the entire environment you created if you add a third `t3a.small` study machine?

188

✓ Correct Answer

Task 4 Conclusion

In this room, you explored the core ideas behind **cloud computing** and how it enables flexible, scalable, and cost-effective access to computing resources.

Key Terminology

Let's quickly review some concepts we learned in this room:

- **Public Cloud**
  Cloud services you access over the internet that many people and companies share.
- **Private Cloud**
  A cloud built just for one company, so they have more control and security.
- **Hybrid Cloud**
  A mix of public and private clouds that can work together and share data.
- **IaaS**
  A service where you rent basic computer parts like servers and storage from the cloud.
- **PaaS**
  A service that gives you a ready-to-use environment to build and run apps without managing servers.
- **SaaS**
  Software you use online without installing anything, like Gmail or Zoom.
- **EC2**
  Amazon's cloud computers that you can quickly create, use, and resize whenever you need them.

We also concluded that the key benefits of cloud computing are:

- Scalability
- On-demand self-service
- Pay only for what you use
- Security
- High availability
- Global access

Further Learning

You are now ready to explore the next key area: Operating Systems. In the Operating Systems Introduction room, you will learn how OS components manage hardware, processes, memory, and security. This knowledge is essential for understanding how cloud systems run applications behind the scenes!

Answer the questions below

I'm ready for the next module!

No answer needed

✓ Correct Answer

Great work Abhishek26! Room completed!

Your skills are skyrocketing!

Completed tasks	Points earned	Streak
4	56	7